Oscillatory activity in the anterior insular cortex of mice: acute effects of a single psychedelic injection

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Introduction

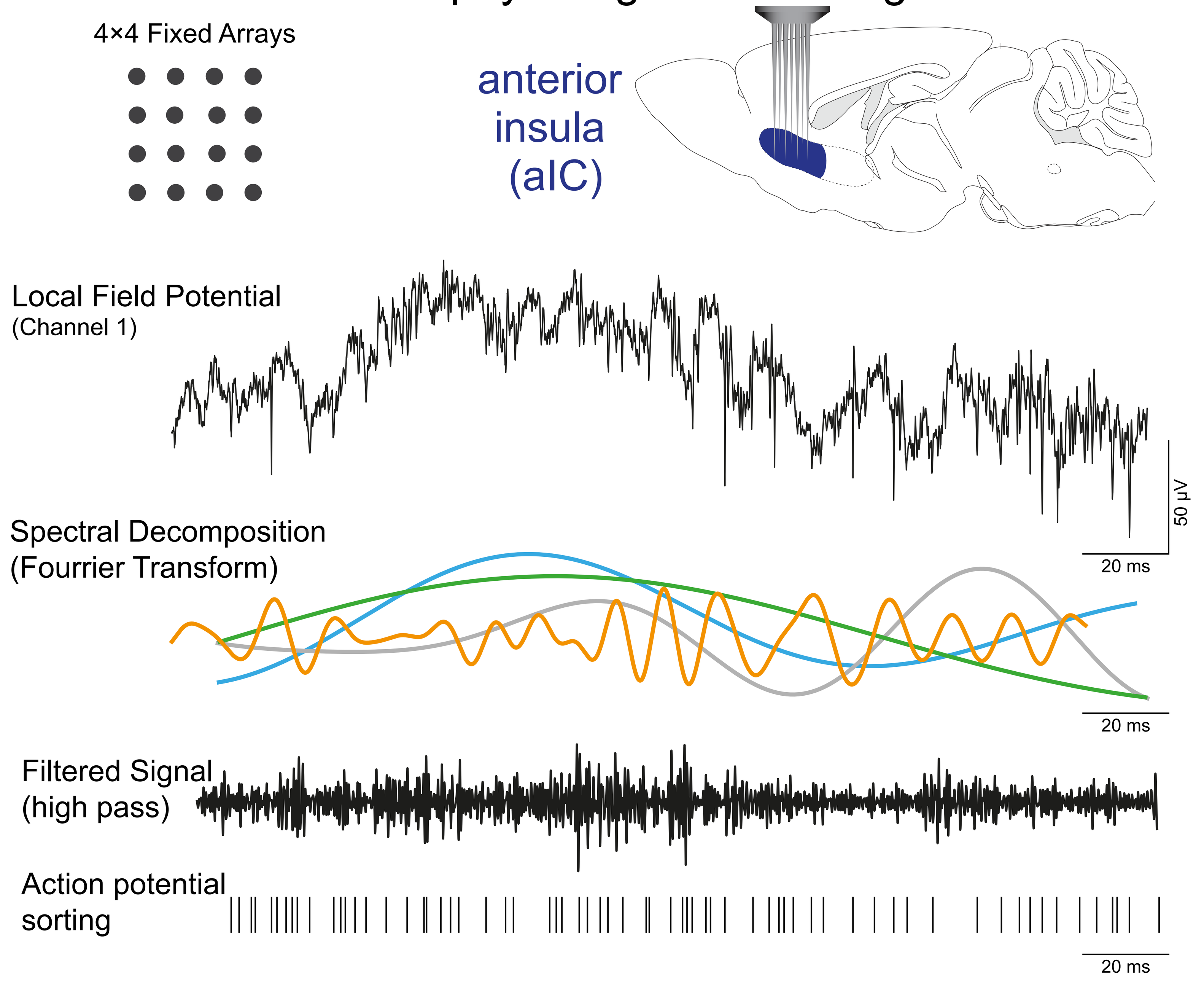
Anxiety disorders affect 5% of Europe's adult population, making them one of the most widespread psychiatric conditions. Psychedelic substances are increasingly employed in the treatment of various psychiatric disorders, including anxiety disorders. These treatments typically involve a single administration and offer lasting benefits ranging from weeks to months. Despite the probable involvement of serotonin 2A receptors (5-HT2A) agonist activity in these drugs, their cellular and temporal effects over days remain unknown.

The insular cortex, also known as the insula, plays a pivotal role in modulating anxiety-related behaviors, with its activity strongly linked to trait anxiety (Nicolas et al., 2023). Notably, this region hosts one of the highest concentrations of 5-HT1A and 5-HT2A receptors in the brain (Ju et al., 2020). Consequently, we propose that the insula may play a significant role in mediating the prolonged anxiolytic effects of 5-HT1A and 5-HT2A activation.

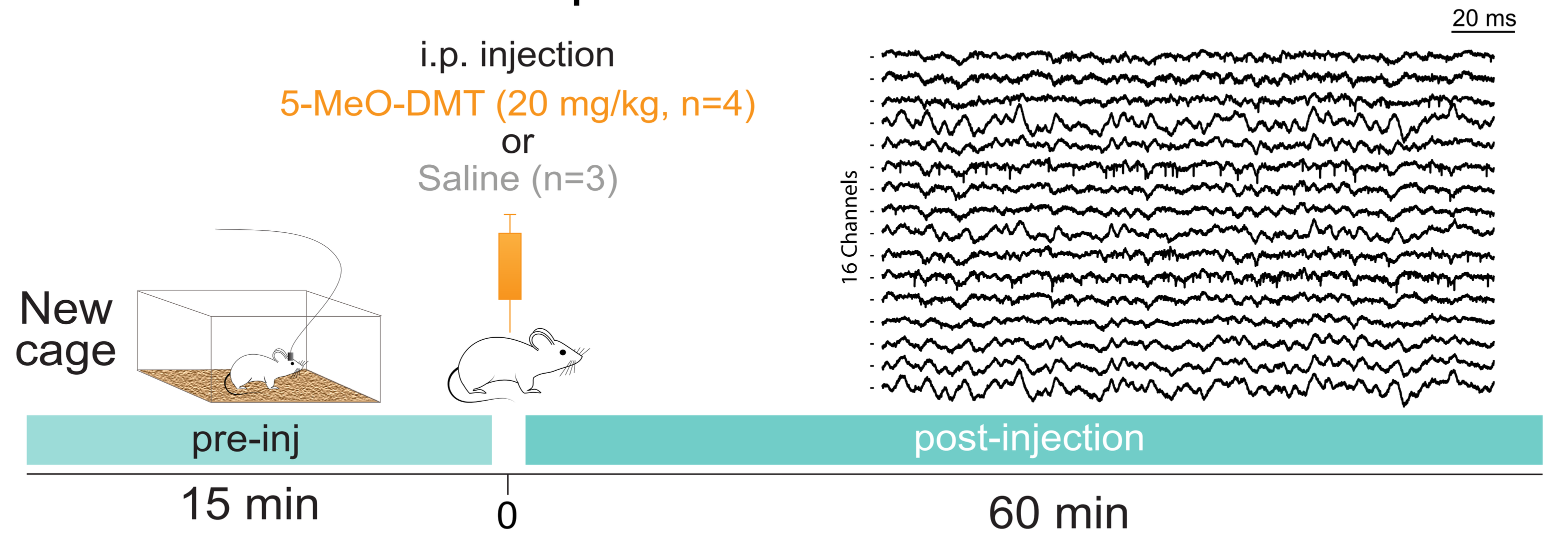
Using in vivo electrophysiological recordings and computational tools, we explore the impact of psychedelics on the function of the anterior insula during anxiety-like behaviors in animal models.

Methods

Electrophysiological recordings

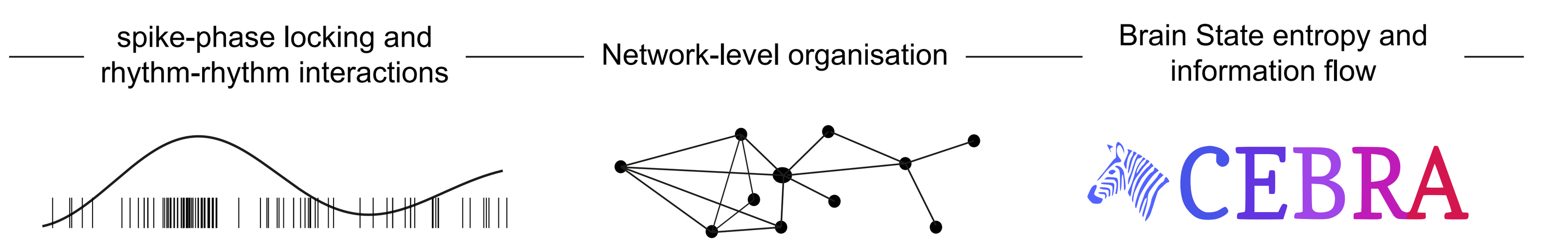


Experimental Timeline

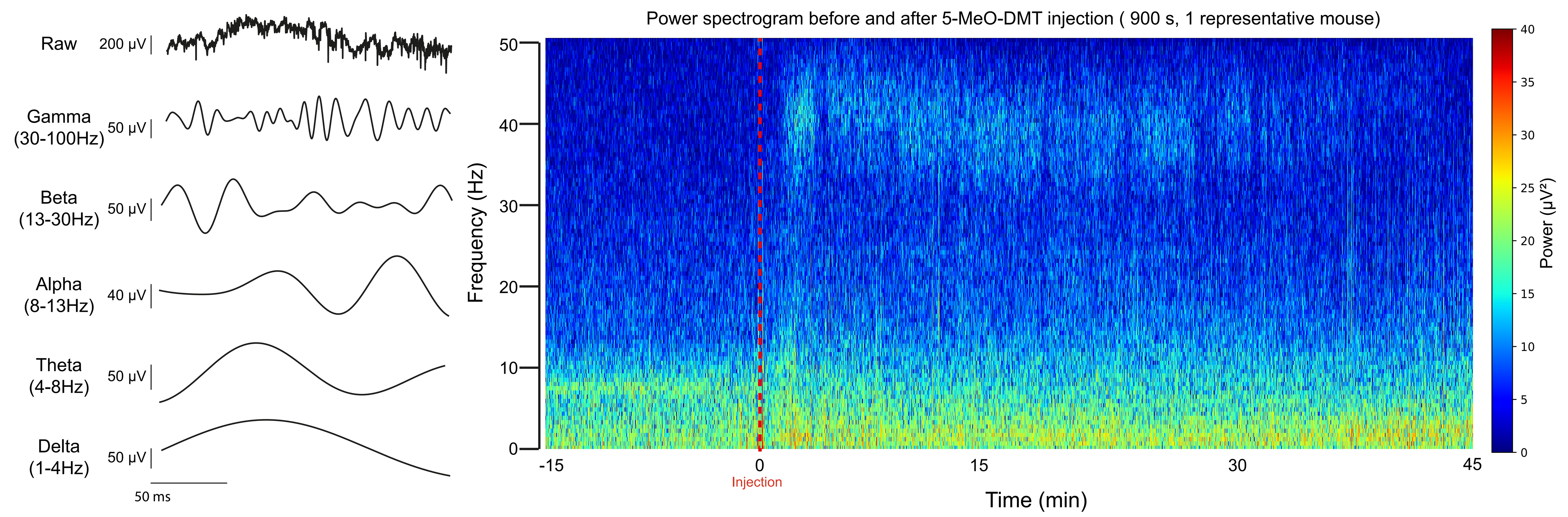


Preliminary Conlcusions and Perspectives

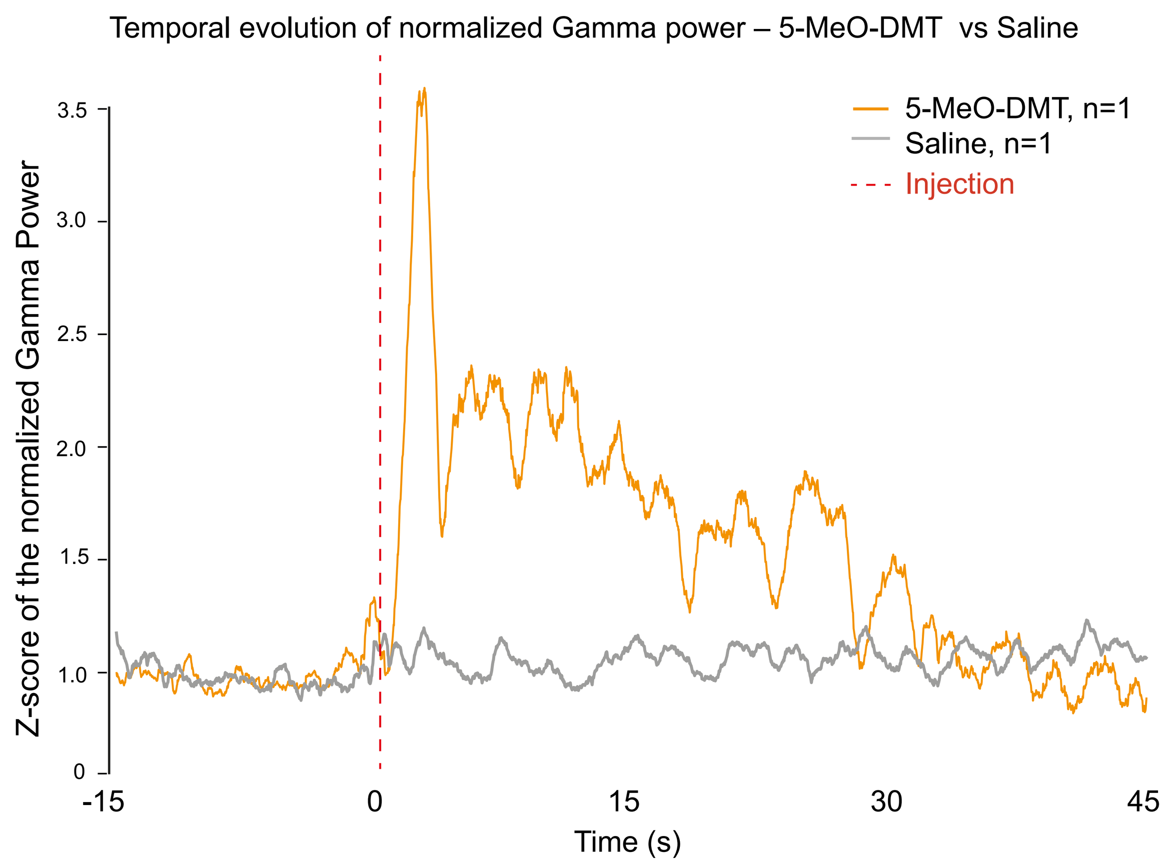
- Increased gamma-band power observed after 5-MeO-DMT injection
- Diverse spike-gamma relatiopnships across neurons with most neurons inhibited
- Overall inhibitory effect of 5-MeO-DMT on average firing rates
- Preliminary cell-type classification
  Stronger **inhibition** in *putative excitatory* neurons
  Stronger **excitation** in *putative inhibitory* neurons

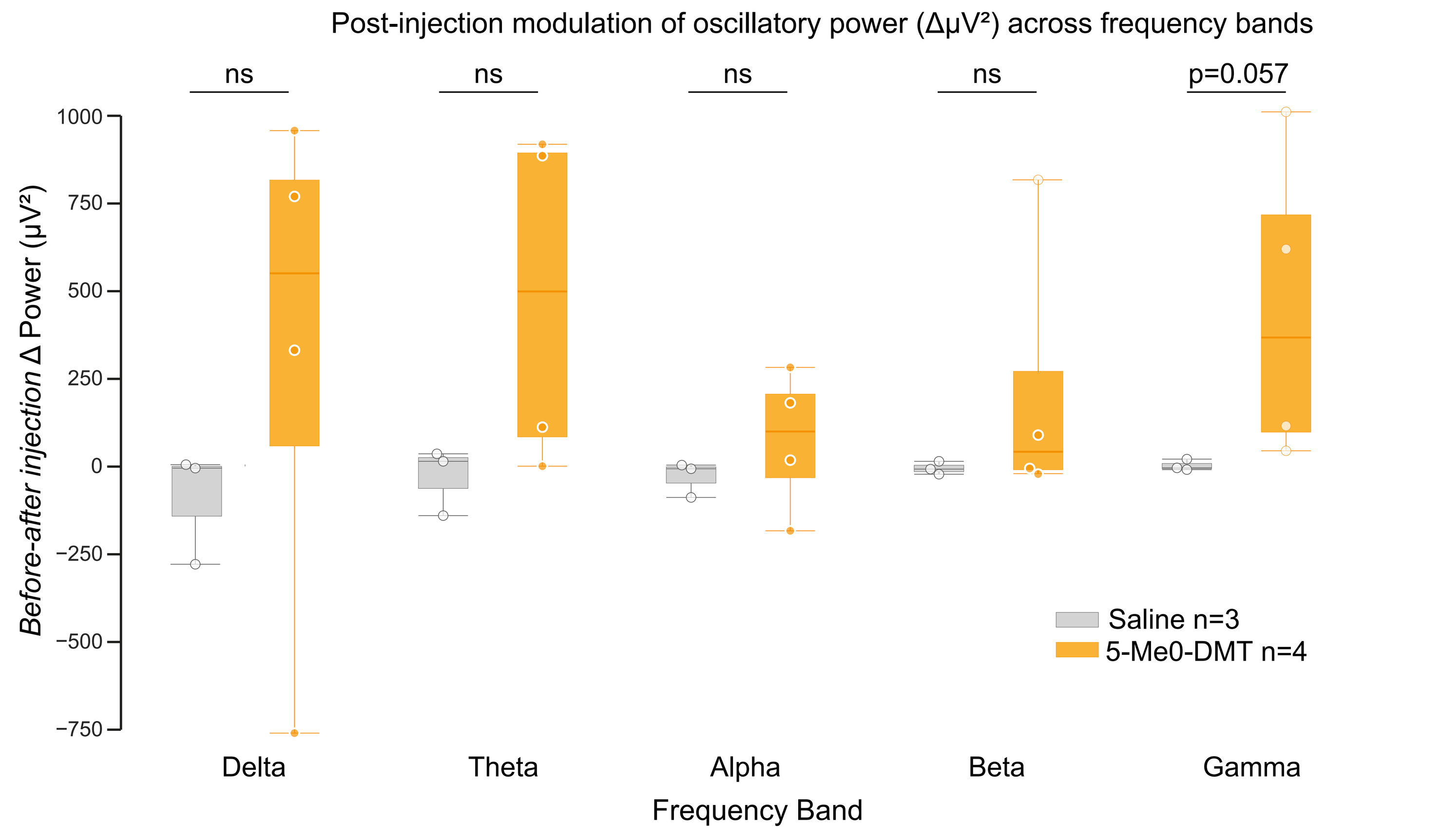


Time-Frequency Profile of 5-MeO-DMT injection

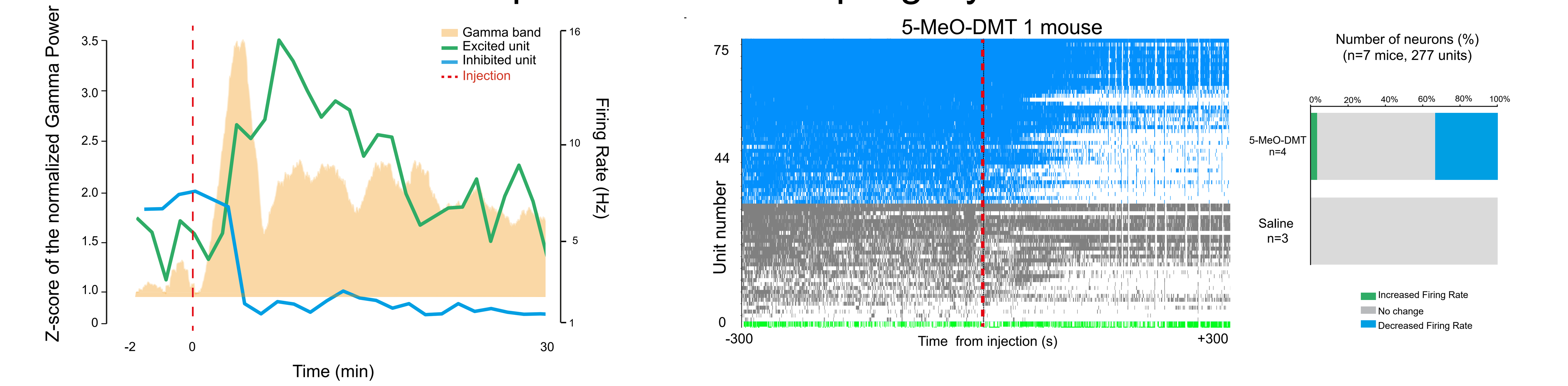


Spectral Power Changes After 5-MeO-DMT

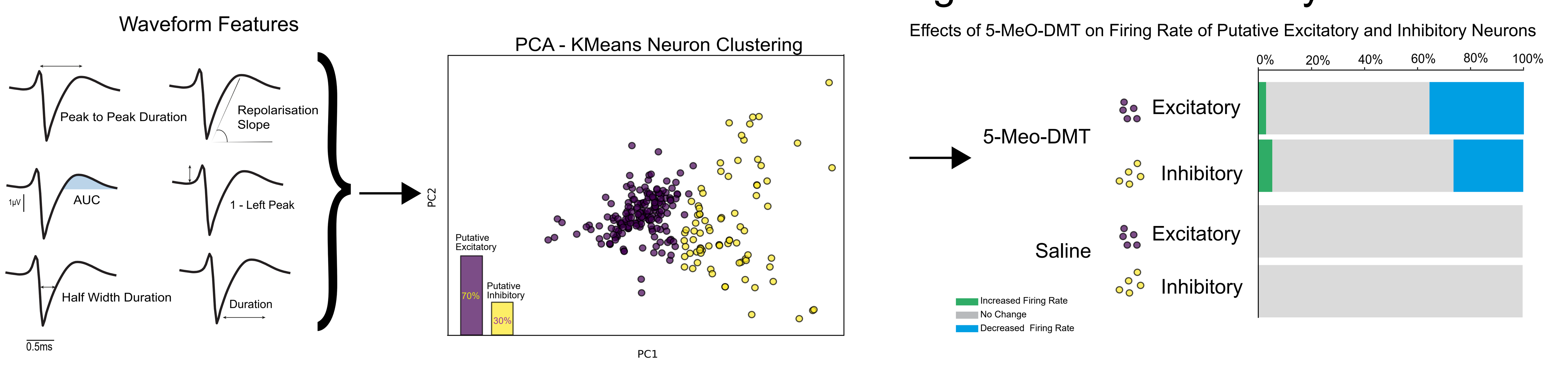




Spike-Gamma Coupling Dynamics



Waveform-Based Neuron Classification and Firing Rate Modulation by 5-MeO-DMT



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anr
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